

Building & Operating a Mini Security Operations Center (SOC)

SOC Implementation and Threat Detection

Team 4 — DEPI 2026

Release 6 | May 2026



Agenda

1. Project Overview & Objectives
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3. VM Inventory & Infrastructure
4. Detection & Monitoring Stack
5. Project Timeline & Milestones
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7. Use Case 1: Brute Force Attack
8. Use Case 2: SQL Injection
9. Use Case 3: Network Scanning
10. Use Cases 4-5: Lateral Movement & Privilege Escalation
11. Risk Assessment & KPIs
12. Testing & Quality Assurance
13. Findings & Recommendations
14. Q&A / Live Demo

Project Overview & Objectives

OVERVIEW:

This project designs, implements, and operates a Mini SOC in a virtualized lab environment simulating a corporate network with Active Directory, a web application (DVWA), workstations, pfSense firewall, Suricata IDS, ModSecurity WAF, and centralized Splunk SIEM.

OBJECTIVE 1:

Simulate a full SOC workflow:

Log Ingestion --> Threat Detection --> Triage --> Reporting

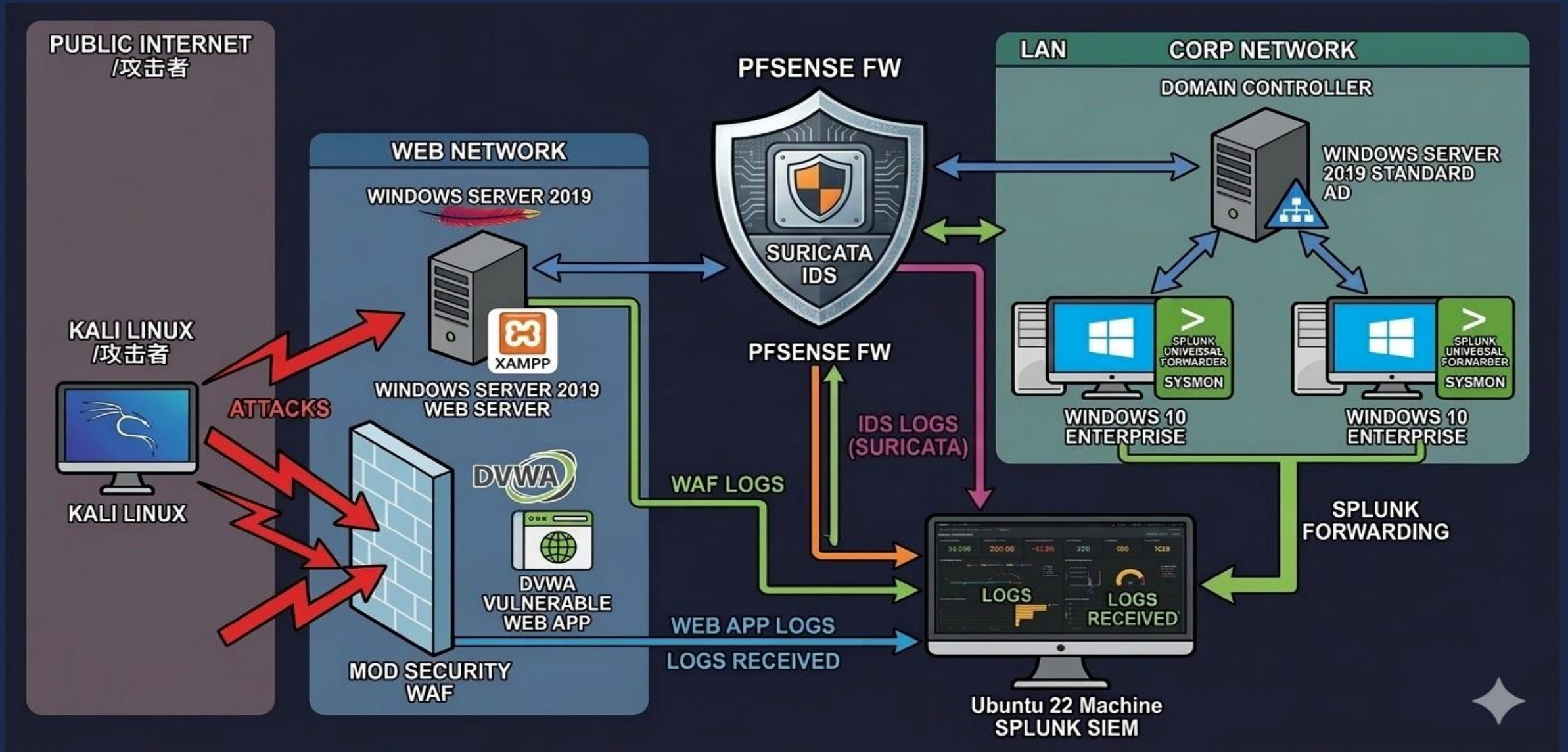
OBJECTIVE 2:

Build a functional SOC using enterprise-grade tools:

- * Splunk Enterprise (SIEM / monitoring)
- * pfSense (firewall / network segmentation)
- * Suricata (IDS/IPS on pfSense)
- * ModSecurity (Web Application Firewall on XAMPP)
- * Sysmon + Universal Forwarders (endpoint visibility)

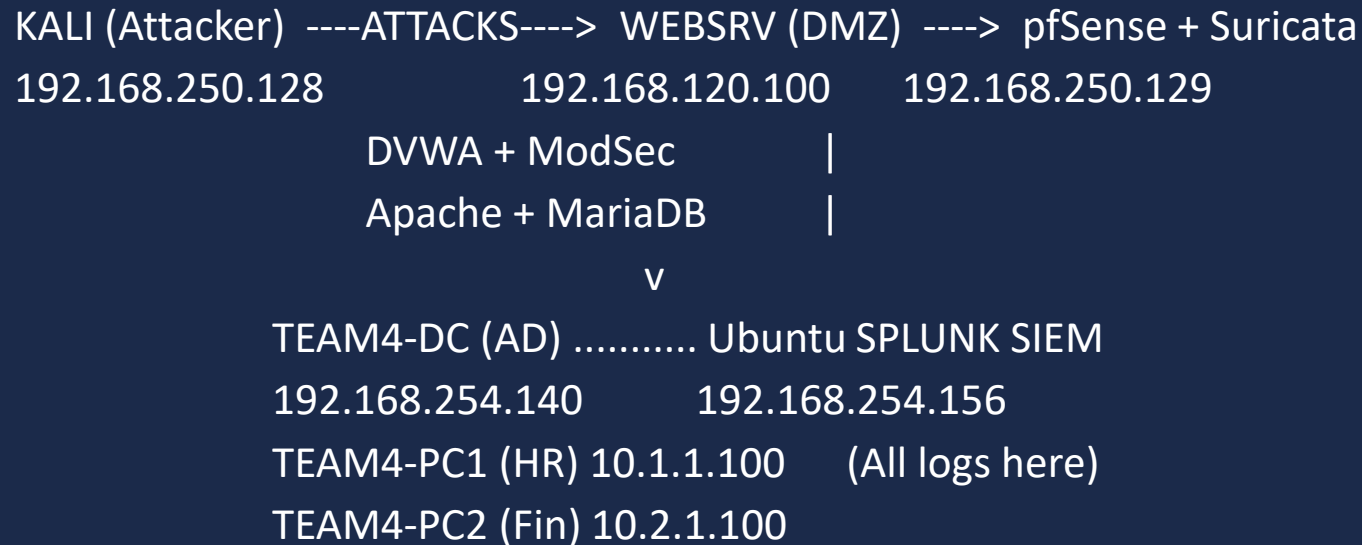
Attacks are simulated from Kali Linux to validate 5 detection use cases.

Lab Architecture & Network Topology



Lab Architecture & Network Topology

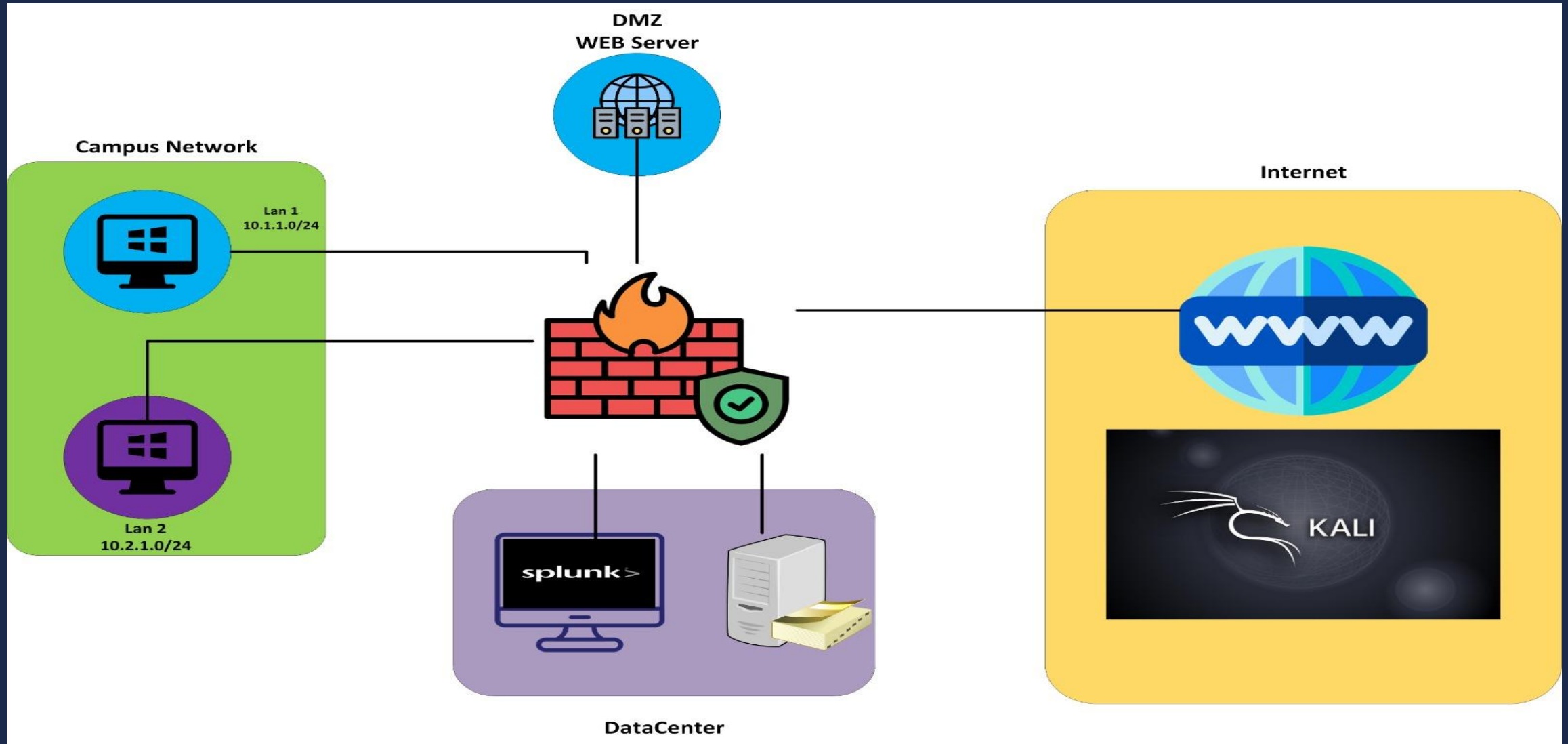
ARCHITECTURE:



NETWORK SEGMENTATION (pfSense — 5 Interfaces):

WAN	192.168.250.0/24	--> Internet / External Exposure	(VMnet0)
LAN	192.168.254.0/24	--> DataCenter / Management	(VMnet1)
DMZ	192.168.120.0/24	--> Web Hosting (WEBSRV)	(VMnet2)
OPT1	10.1.1.0/24	--> Corporate HR Department	(VMnet3)
OPT2	10.2.1.0/24	--> Corporate Finance Department	(VMnet4)

Lab Architecture & Network Topology



Infrastructure: VMs & Credentials

Hostname	OS	IP Address	Role	Domain
TEAM4-DC	Win Server 2019	192.168.254.140	Domain Controller (AD/DNS)	SOC.DEPI
WEBSRV	Win Server 2019	192.168.120.100	XAMPP + DVWA + MariaDB	SOC.DEPI
TEAM4-PC1	Win 10 Ent.	10.1.1.100	HR Workstation	SOC.DEPI
TEAM4-PC2	Win 10 Ent.	10.2.1.100	Finance Workstation	SOC.DEPI
pfsense	freebsd:11.2	192.168.254.2 192.168.120.1 10.1.1.1 10.2.1.1 192.168.250.129 192.168.120.1	FW	Non-Domain
UBUNTU	UBUNTU 22.04	192.168.254.156	SPLUNK SIEM	Non-Domain
KALI	Kali 2025.4	192.168.250.128	Attacker (VAPT)	Non-Domain

Detection Stack & Splunk Index Map

Layer	Tool	Source / Index	Purpose
Layer 1	ModSecurity WAF	waf	Detect SQLi, XSS on WEBSRV
Layer 2	Suricata IDS/IPS	suricata	Network intrusion detection on pfSense
Layer 3	pfSense Filterlog	pfsense_fw	Firewall traffic logging (allow/deny)
Layer 4	Apache Access Log	webapp	HTTP request logging on WEBSRV
Layer 5	Sysmon + WinEventLog	windows_dc	DC process/logon/network events
Layer 6	Sysmon (UF)	windows_pc1	HR workstation endpoint telemetry
Layer 7	Sysmon (UF)	windows_pc2	Finance workstation endpoint telemetry
Layer 8	Sysmon (UF)	windows_websrv	Web server host-level events

Timeline, Milestones & Team Assignment

4-WEEK SPRINT:

- Week 1 (Mar): Splunk + pfSense + DC setup --> Log pipeline operational
- Week 2 (Apr): UF + Sysmon + Suricata + ModSec --> Correlation rules documented
- Week 3 (May): SPL rules + Attack simulations --> Triage sheets + IOCs
- Week 4 (May): IR Playbook + Report + Demo --> Final submission

MILESTONES:

- M1 (Jan 27) Project Proposal
- M2 (Mar 1) Literature Review
- M3 (Apr 3) System Design
- M4 (May 17) Functional SOC
- M5 (May 24) Final Report + Presentation

TEAM ASSIGNMENT:

- Abdalaziz Mohammad --> AD, Sysmon, GPO, Windows Logging
- Ahmed Hamdy --> Splunk Installation
- Marko Maged --> pfSense Firewall
- Ahmed Atef --> Suricata IDS/IPS
- Bassam Alsayed --> XAMPP Web Server + DVWA
- Ahmed Tarek --> ModSecurity WAF

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Timeline, Milestones & Team Assignment

Team4 Project Installing Sequence

installing windows server 2019 and AD

installing first windows 10 enterprise machine

installing second windows 10 enterprise machine

install splunk on ubuntu

AD Environment CONFIGURATIONS

installing Splunk Forwarder on THE AD machine

installing Splunk Forwarder on THE First WINDOWS 10 ENTERPRISE machine

installing Splunk Forwarder on THE SECOND WINDOWS 10 ENTERPRISE machine

installing splunk forwarder on windows Server 2019 for XAMP and DVAW machines

Preparing AD machine Before installing Splunk Forwarder

Preparing Splunk machine to receive from Splunk Forwarder

Preparing THE First WINDOWS 10 ENTERPRISE machine before installing Splunk Forwarder

Preparing THE Second WINDOWS 10 ENTERPRISE machine before installing Splunk Forwarder

Preparing windows Server 2019 for XAMP and DVAW machines Before installing splunk forwarder

installing splunk forwarder FOR WEB APP. AND WAF on windows Server 2019 for XAMP and DVAW machines

installing modsecurity WAF on windows Server 2019 for XAMP and DVAW machine

Timeline, Milestones & Team Assignment

installing kali linux

installing SYSMON on THE AD machine and configure splunk to receive it

installing SYSMON on THE FIRST WINDOWS 10 ENTERPRISE machine and configure Splunk to receive from it

installing SYSMON on THE Second WINDOWS 10 ENTERPRISE machine and configure Splunk to receive from it

installing XAMP Web Server and DVWA WEB App. on windows Server 2019 for machine

AD Environment CONFIGURATIONS for Preparing to The Simulation of The Attacks

installing SYSMON on windows 2019 WITH XAMP AND DVWA and configure splunk to receive it

installing pfsense FW and segmentation

Config. Pfsense FW RULES and send FW log to splunk part1

Config. Pfsense FW RULES and send FW log to splunk part2

installing suricata and config rules and send log to splunk

reinstalling splunk and configure indexes and data sources

pfsense FW Rules for INTER-VLAN routing

Development the SOC lab by adding another HOST

team4- project documentation release 4

pfsense use case

suricata use case

NETWORK SCANNING USE CASE attacking dvwa we app

Use Case 1: Brute Force Attack

Description: Multiple failed authentication attempts from a single source IP against RDP/SSH/Web.

Why It Matters: Leads to account compromise, data breaches, and ransomware deployment.

Data Sources: Windows EventCode 4625 (Failed Logon), /var/log/secure, HTTP 401 status codes.

MITRE ATT&CK: T1110 — Brute Force

Payload:

```
hydra -L /usr/share/wordlists/metasploit/namelist.txt -p WrongPass123 192.168.254.140 rdp
```

OR

```
for i in {1..20}; do net rdp share -I 192.168.254.140 -U fakeuser%wrongpass; done
```

SPL:

```
index="windows_dc" EventCode=4625
| stats count min(_time) as firstTime
    max(_time) as lastTime
    by Source_Network_Address, Account_Name
| where count >= 5
| eval severity="high"
```



- My Computer
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 - Windows 10 x64 pc1
 - Windows 10 x64 pc2
 - kali-linux-2025.4-vmware-amd64 for VAPT
 - Windows Server 2019 for XAMP and DVWA
 - Pfsense FW
 - Ubuntu 64-bit For Splunk

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ping 192.168.254.140  
PING 192.168.254.140 (192.168.254.140) 56(84) bytes of data.  
64 bytes from 192.168.254.140: icmp_seq=1 ttl=127 time=2.72 ms  
64 bytes from 192.168.254.140: icmp_seq=2 ttl=127 time=2.54 ms  
64 bytes from 192.168.254.140: icmp_seq=3 ttl=127 time=1.09 ms  
64 bytes from 192.168.254.140: icmp_seq=4 ttl=127 time=270 ms  
64 bytes from 192.168.254.140: icmp_seq=5 ttl=127 time=2.59 ms  
^C  
--- 192.168.254.140 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4009ms  
rtt min/avg/max/mdev = 1.087/55.742/269.780/107.020 ms  
(kali@kali)-[~]  
$ hydra -L /usr/share/wordlists/metasploit/namelist.txt -p WrongPass123 192.168.254.140 rdp  
  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-16 07:16:27  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1909 login tries (l:1909/p:1), ~478 tries per task  
[DATA] attacking rdp://192.168.254.140:3389/  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[STATUS] 90.00 tries/min, 90 tries in 00:01h, 1820 to do in 00:21h, 3 active  
[ERROR] freerdp: The connection failed to establish.  
[STATUS] 88.67 tries/min, 266 tries in 00:03h, 1644 to do in 00:19h, 3 active  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[STATUS] 87.29 tries/min, 611 tries in 00:07h, 1299 to do in 00:15h, 3 active  
[ERROR] freerdp: The connection failed to establish.  
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.  
(kali@kali)-[~]  
$
```

Ubuntu 64-bit For Splunk - VMware Workstation

FileEditViewVMTabsHelp

Library

My Computer

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HomeUbuntu 64-bit For SplunkPfsense FWWindows 10 x64 pc1Windows Server 2019 ADWindows 10 x64 pc2Windows Server 2019 for XAMP...kali-lin...

ActivitiesFirefox Web Browser13:59 16 مای

Search | Splunk 10.2.0

pfsense.SOC.DEPI - Statu

testmyids.com/

localhost:8000/en-US/app/search/search?q=search index%3D"windows_dc" EventCode%3D4625%0A| stats count min(

90%

splunk>enterprise

SearchAnalyticsDatasetsReportsAlertsDashboardsModules

New Search

index="windows_dc" EventCode=4625

| stats count min(_time) as firstTime

max(_time) as lastTime

by Source_Network_Address, Account_Name

| where count >= 5

| eval severity="high"

1,230 events (5/15/26 1:00:00.000 PM to 5/16/26 1:59:47.000 PM) No Event Sampling

Events (1,230)PatternsStatistics (8)Visualization

Show: 20 Per PageFormatPreview: On

Source_Network_Address	Account_Name	count	firstTime	lastTime	severity
192.168.250.128	-	1230	1778885005.409	1778929185.395	high
192.168.250.128	10	6	1778885009.016	1778929162.677	high
192.168.250.128	12	6	1778885007.478	1778929162.742	high
192.168.250.128	14	5	1778885011.042	1778929164.706	high
192.168.250.128	15	6	1778885011.271	1778929164.755	high
192.168.250.128	16	5	1778885011.528	1778888414.196	high
192.168.250.128	2	5	1778885013.564	1778929167.049	high
192.168.250.128	4	5	1778885015.586	1778929169.080	high

27°C

شمس

Search

01:59 م

2026/05/16

Use Case 2: SQL Injection (SQLi)

Description: Malicious SQL code inserted into URLs/forms to manipulate the database (OWASP Top 10 #1).

Why It Matters: Data exfiltration of PII/financial data, authentication bypass, Remote Code Execution.

Data Sources: Apache access logs (webapp index), ModSecurity WAF logs (waf index).

MITRE ATT&CK: T1190 — Exploit Public-Facing Application

Payload:

```
1' UNION SELECT user, password FROM users #
```

OR

```
' OR '1'='1
```

OR

```
sqlmap -u http://192.168.120.100/dvwa/vulnerabilities/sqli//?id=1&Submit=Submit
```

SPL:

```
index=* "UNION"
```

```
| stats values(sourcetype) as "Source Types",  
    count by host
```

```
-- Alternative (WAF + Web combined) --
```

```
(index=waf OR index=webapp) ("UNION SELECT" OR "OR 1=1")
```

```
| rex field=_raw "GET (?<url>.+?) HTTP"
```

```
| table _time host url
```



DVWA Security

Submit

```
ID: 1' UNION SELECT user, password FROM users #
First name: smithy
Surname: 5f4dcc3b5aa765d61d8327deb882cf99
```

- https://en.wikipedia.org/wiki/SQL_injection
- <https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/>
- https://owasp.org/www-community/attacks/SQL_injection
- <https://bobby-tables.com/>



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Vulnerability: SQL Injection

User ID:

ID: ' OR '1'='1
First name: admin
Surname: admin

ID: ' OR '1'='1
First name: Gordon
Surname: Brown

ID: ' OR '1'='1
First name: Hack
Surname: Me

ID: ' OR '1'='1
First name: Pablo
Surname: Picasso

ID: ' OR '1'='1
First name: Bob
Surname: Smith

More Information

- https://en.wikipedia.org/wiki/SQL_injection
- <https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/>
- https://owasp.org/www-community/attacks/SQL_injection
- <https://bobby-tables.com/>

Home

Instructions

Setup / Reset DB

Brute Force

Command Injection

CSRF

File Inclusion

File Upload

Insecure CAPTCHA

SQL Injection

SQL Injection (Blind)

Weak Session IDs

XSS (DOM)

XSS (Reflected)

XSS (Stored)

CSP Bypass

JavaScript Attacks

Authorisation Bypass

Open HTTP Redirect

Cryptography

API

DVWA Security

Library

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Home Ubuntu 64-bit For Splunk Pfsense FW Windows 10 x64 pc1 Windows Server 2019 AD Windows 10 x64 pc2 Windows Server 2019 for XAMP... kali-li...

Session Actions Edit View Help

got a 302 redirect to 'http://192.168.120.100/dvwa/login.php'. Do you want to follow? [Y/n] y

[07:53:47] [CRITICAL] WAF/IPS identified as 'ModSecurity (Trustwave)'

you have not declared cookie(s), while server wants to set its own ('security=low;PHPSESSID=unafi5eb2d7 ... gnbajmqeie'). Do you want to use those [Y/n] y

[07:53:51] [INFO] testing if the target URL content is stable

[07:53:51] [WARNING] GET parameter 'id' does not appear to be dynamic

[07:53:51] [WARNING] heuristic (basic) test shows that GET parameter 'id' might not be injectable

[07:53:52] [INFO] testing for SQL injection on GET parameter 'id'

[07:53:52] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'

[07:53:52] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'

[07:53:52] [INFO] testing 'MySQL >= 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'

[07:53:52] [INFO] testing 'PostgreSQL AND error-based - WHERE or HAVING clause'

[07:53:52] [INFO] testing 'Microsoft SQL Server/Sybase AND error-based - WHERE or HAVING clause (IN)'

[07:53:52] [INFO] testing 'Oracle AND error-based - WHERE or HAVING clause (XMLType)'

[07:53:52] [INFO] testing 'Generic inline queries'

[07:53:52] [INFO] testing 'PostgreSQL > 8.1 stacked queries (comment)'

[07:53:52] [INFO] testing 'Microsoft SQL Server/Sybase stacked queries (comment)'

[07:53:52] [INFO] testing 'Oracle stacked queries (DBMS_PIPE.RECEIVE_MESSAGE - comment)'

[07:53:53] [INFO] testing 'MySQL >= 5.0.12 AND time-based blind (query SLEEP)'

[07:53:53] [INFO] testing 'PostgreSQL > 8.1 AND time-based blind'

[07:53:53] [INFO] testing 'Microsoft SQL Server/Sybase time-based blind (IF)'

[07:53:53] [INFO] testing 'Oracle AND time-based blind'

it is recommended to perform only basic UNION tests if there is not at least one other (potential) technique found. Do you want to reduce the number of requests? [Y/n] y

[07:53:55] [INFO] testing 'Generic UNION query (NULL) - 1 to 10 columns'

[07:53:55] [WARNING] GET parameter 'id' does not seem to be injectable

[07:53:55] [WARNING] GET parameter 'Submit' does not appear to be dynamic

[07:53:55] [WARNING] heuristic (basic) test shows that GET parameter 'Submit' might not be injectable

[07:53:55] [INFO] testing for SQL injection on GET parameter 'Submit'

[07:53:55] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'

[07:53:56] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'

[07:53:56] [INFO] testing 'MySQL >= 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'

[07:53:56] [INFO] testing 'PostgreSQL AND error-based - WHERE or HAVING clause'

[07:53:56] [INFO] testing 'Microsoft SQL Server/Sybase AND error-based - WHERE or HAVING clause (IN)'

[07:53:56] [INFO] testing 'Oracle AND error-based - WHERE or HAVING clause (XMLType)'

[07:53:56] [INFO] testing 'Generic inline queries'

[07:53:56] [INFO] testing 'PostgreSQL > 8.1 stacked queries (comment)'

[07:53:57] [INFO] testing 'Microsoft SQL Server/Sybase stacked queries (comment)'

[07:53:57] [INFO] testing 'Oracle stacked queries (DBMS_PIPE.RECEIVE_MESSAGE - comment)'

[07:53:57] [INFO] testing 'MySQL >= 5.0.12 AND time-based blind (query SLEEP)'

[07:53:57] [INFO] testing 'PostgreSQL > 8.1 AND time-based blind'

[07:53:57] [INFO] testing 'Microsoft SQL Server/Sybase time-based blind (IF)'

[07:53:57] [INFO] testing 'Oracle AND time-based blind'

[07:53:57] [INFO] testing 'Generic UNION query (NULL) - 1 to 10 columns'

[07:53:57] [WARNING] GET parameter 'Submit' does not seem to be injectable

[07:53:57] [CRITICAL] all tested parameters do not appear to be injectable. Try to increase values for '--level/'--risk' options if you wish to perform more tests. If you suspect that there is some kind of protection mechanism involved (e.g. WAF) maybe you could try to use option '--tamper' (e.g. '--tamper=space2comment') and/or switch '--random-agent'

[*] ending @ 07:53:57 /2026-05-16/

DVWA Security

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

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Activities Firefox Web Browser 15:19 16 م

Home Ubuntu 64-bit For Splunk Pfsense FW Windows 10 x64 pc1 Windows Server 2019 AD Windows 10 x64 pc2 Windows Server 2019 for XAMP... kali-lin...

Dashboards | Splunk 10.2 WORKING DASHBOARD pfSense.SOC.DEPI - Statu testmyids.com/

localhost:8000/en-US/app/search/working_dashboard_panels 90%

No results found.

PANEL 3 — SQLI ATTACKS

PANEL 4 — DNS RECON

No results found.

FileEditViewVMTabsHelp

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Activities

Firefox Web Browser

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Home

Ubuntu 64-bit For Splunk

Pfsense FW

Windows 10 x64 pc1

Windows Server 2019 AD

Windows 10 x64 pc2

Windows Server 2019 for XAMP...

kali-lin...

Firefox Web Browser

15:18 16 مای

Dashboards | Splunk 10.2

Triggered Alerts

pfSense.SOC.DEPI - Statu

testmyids.com/

localhost:8000/en-US/app/search/triggered_alerts?eai%3Aacl.app=search&eai%3Aacl.owner=*&severity=*&alert_name=90%

splunk>enterprise

Apps

Administrator

6 Messages

Settings

Activity

Help

Find

Triggered Alerts

Filter

AppSearch & Report...

OwnerAll owners

SeverityAll severity

Alert nameAll alerts

Showing 1 - 20 of 5198 results

Show: 20 per p...

1of 260 pages

	Time	Alert name	App	Type	Severity	Mode	Actions
☐	2026-05-16 15:16:15 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:16:10 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:15:49 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:15:44 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:15:43 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:15:41 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:15:00 EEST	Brute Force Detection2 - Failed Logons	search	Scheduled	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:13:03 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:12:58 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:12:46 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:12:41 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:12:41 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:12:38 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:10:58 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:10:53 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:10:33 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:10:18 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:10:13 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:10:12 EEST	SQL Injection (Works on your old DVWA attack)	search	Real-time	High	Per Result	View Results Edit Alert Delete

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Use Case 3: Network Scanning (Reconnaissance)

Description: Mapping live hosts, open ports, and running services using Nmap — Cyber Kill Chain Phase 1.

Why It Matters: High-fidelity precursor to exploitation; early detection enables hardening.

Data Sources: pfSense firewall logs (filterlog), Suricata IDS alerts.

MITRE ATT&CK: T1046 — Network Service Discovery | T1595 — Active Scanning

Payload:

```
nmap -Pn -sS -A -p- -T4 192.168.254.140
nmap -Pn -sS -A -p- -T4 10.1.1.100
nmap -Pn -sS -A -p- -T4 10.2.1.100
nmap -Pn -sS -A -p- -T4 192.168.120.100
nmap -Pn -sS -A -p- -T4 192.168.250.129
```

SPL:

```
index=pfsense_fw
| stats dc(dest_port) as scanned_ports
      count by src_ip dest_ip
| where scanned_ports > 20
| sort - scanned_ports
```

OR

```
index=pfsense_fw src_ip=192.168.250.128
| stats dc(dest_port) as unique_ports
      values(dest_port) as ports
      by src_ip dest_ip
| where unique_ports > 15
| sort - unique_ports
```


Ubuntu 64-bit For Splunk - VMware Workstation

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- Windows Server 2019 for XAMP and DVWA
- Pfsense FW
- Ubuntu 64-bit For Splunk

HomeUbuntu 64-bit For SplunkPfsense FWWindows 10 x64 pc1Windows Server 2019 ADWindows 10 x64 pc2Windows Server 2019 for XAMP...kali-lin...

ActivitiesFirefox Web Browser15:56 16 مای

Search | Splunk 10.2.0

pfsense.SOC.DEPI - Statu

testmyids.com/

localhost:8000/en-US/app/search/search?s=%2FservicesNS%2Fadministrator%2Fsearch%2Fsaved%2Fsearches%2FAlert190%

splunk>enterpriseAppsAdministrator6 MessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboardsModulesSearch & Reporting

Alert 1 — Port Scan DetectionSPLConvert to SPL2SaveSave AsViewCreate Table ViewClose

index=pfsense_fw src_ip=192.168.250.128
| stats dc(dest_port) as unique_ports
values(dest_port) as ports
by src_ip dest_ip
| where unique_ports > 15
| sort - unique_ports

Time range: All time

1,026,448 events (before 5/16/26 3:54:43.000 PM)No Event SamplingJobPauseRefreshDownloadVerbose Mode

Events (1,026,448)PatternsStatistics (5)Visualization

Show: 20 Per PageFormatPreview: On

src_ip	dest_ip	unique_ports	ports
192.168.250.128	192.168.250.129	65530	1 10 100 1000 10000 10001 10002 10003 10004 10005 10006 10007 10008 10009 1001 10010 10011 10012 10013 10014 10015

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Activities

Firefox Web Browser

Triggers Alerts

pfSense.SOC.DEPI - Statu

testmyids.com/

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localhost:8000/en-US/app/search/triggered_alerts?eai%3Aacl.app=search&eai%3Aacl.owner=*&severity=*&alert_name=90%

splunk>enterprise

Administrator6 MessagesSettingsActivityHelpFind

Triggered Alerts

Filter

AppSearch & Report...OwnerAll ownersSeverityAll severityAlert nameAll alerts

Showing 1 - 20 of 5468 results

Show: 20 per p...1of 274 pages

☐	Time	Alert name	App	Type	Severity	Mode	Actions
☐	2026-05-16 15:36:31 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:29 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:29 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:28 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:27 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:27 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:26 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:26 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:25 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:25 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:24 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:24 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:23 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:22 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:21 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:20 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:20 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:19 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete
☐	2026-05-16 15:36:19 EEST	Alert 1 — Port Scan Detection	search	Real-time	High	Per Result	View Results Edit Alert Delete

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Firefox Web Browser

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Dashboards | Splunk 10.2 x Nmap Reconnaissance M x pfSense.SOC.DEPI - Statu x testmyids.com/

localhost:8000/en-US/app/search/nmap_reconnaissance_monitoring?tab=layout_1&form.global_time.earliest=0&form.g 90%

splunk>enterprise Apps

Search Analytics Datasets Reports Alerts Dashboards Modules

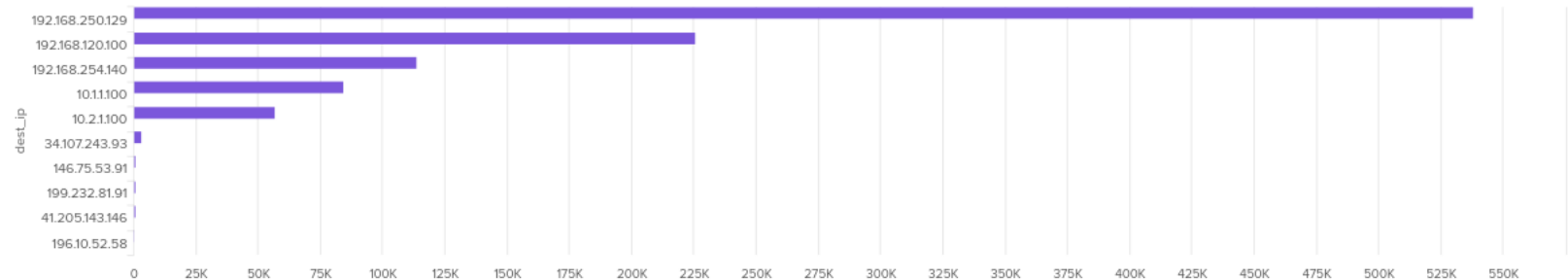
Nmap Reconnaissance Monitoring

Nmap Reconnaissance Monitoring

Global Time Range

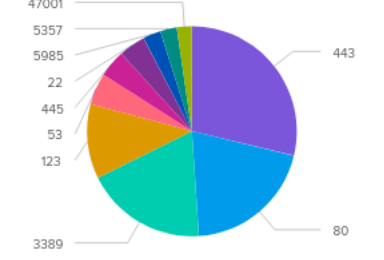
All time

Panel 1 — Top Scanned Hosts



dest_ip	count	percent
192.168.250.129	540	100%
192.168.120.100	225	41.67%
192.168.254.140	110	20.37%
10.11.100	85	15.74%
10.2.1.100	60	11.11%
34.107.243.93	5	0.93%
146.75.53.91	2	0.37%
199.232.81.91	1	0.19%
41.205.143.146	1	0.19%
196.10.52.58	1	0.19%

Panel 2 — Top Targeted Ports



port	count	percent
443	443	100%
80	80	18.06%
3389	3389	76.41%
47001	47001	10.51%
5357	5357	1.20%
5985	5985	1.35%
22	22	0.50%
445	445	1.01%
53	53	0.12%
123	123	0.28%

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Activities

Firefox Web Browser

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localhost:8000/en-US/app/search/nmap_reconnaissance_monitoring?tab=layout_1&form.global_time.earliest=0&form.g

90%

splunk>enterprise

SearchAnalyticsDatasetsReportsAlertsDashboardsModules

Nmap Reconnaissance Monitoring

Nmap Reconnaissance Monitoring

Global Time Range

All time

Panel 3 — Scan Timeline

Panel 4 — AD Recon Activity

count

dest_port

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

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2026/05/16

Risk Assessment & Key Performance Indicators

RISK ASSESSMENT:

Risk	Impact	Mitigation
Resource Exhaustion	High	Limit Win10 VMs to 2GB RAM each
IP Address Conflicts	Medium	Strict IP standard; never change subnets
Log Injection Failure	High	Verify UF connectivity via ping + telnet 9997
WAF False Positives	Medium	Run ModSecurity in Detection-Only first

KEY PERFORMANCE INDICATORS (KPIs):

- 99% System Uptime
Splunk web interface availability over 4-week test phase
- 100% Log Coverage
All critical hosts (TEAM4-DC, WEBSRV, pfSense) forwarding successfully
- 0 False Negatives
Zero missed detections during SQLi and NTLM relay simulations
- < 2 min Response Time
Alert generation in Splunk within 2 minutes of trigger event

Testing & Quality Assurance

Test ID	Scenario	Expected Outcome	Status
TC-01	Brute Force from Kali (192.168.250.128)	Splunk alert after >= 5 failed logons	PASSED
TC-02	SQL Injection on DVWA (UNION SELECT)	ModSecurity logs + Splunk dashboard shows UNION	PASSED
TC-03	Nmap Scan (-sV -A -p-) via WAN	pfSense/Suricata logs traffic; Splunk alert fires	PASSED

Findings & Recommendations

FINDINGS:

- * i5-1235U / 16GB RAM successfully handles the complete 7-VM lab
- * All 5 use cases detected and validated with zero false negatives
- * Multi-layer detection (WAF + IDS + Endpoint + SIEM) works as designed
- * pfSense network segmentation effectively isolates attack paths
- * Splunk provides powerful correlation across all log sources

RECOMMENDATIONS:

- * Upgrade to 32GB RAM for scaling beyond 10 use cases
- * Add dedicated Syslog-ng server for Linux log scalability
- * Integrate SOAR platform for automated IP blocking
(pfSense API triggered by Splunk high-severity alerts)
- * Add threat intelligence feeds (AlienVault OTX, Tor exit nodes)
for IOC enrichment in Splunk lookups
- * Transition ModSecurity from Detection-Only to Enforcement mode

GRADING CRITERIA:

- 40% Technical Implementation (Splunk/pfSense configuration)
- 30% Testing (Attack simulations and validation)
- 20% Documentation (Report, playbooks, architecture docs)
- 10% Presentation

Incident Response (IR) Phase

Incident Response Framework:

The SOC IR Lifecycle (NIST SP 800-61)

- **Preparation:** Hardening pfSense, configuring Suricata rules, and deploying Splunk Universal Forwarders.
- **Detection & Analysis:** Monitoring the "Security Posture" dashboard. Identifying Nmap scans (recon) and SQLi payloads (exploitation).
- **Containment:** Using pfSense "EasyRule" to block Attacker IP (192.168.250.128) at the WAN interface.
- **Post-Incident:** Reviewing logs to close the vulnerability in DVWA or reset DC passwords.
- **Visual:** A circular diagram showing the 4 phases of NIST IR.

Incident Response (IR) Phase

Incident Response Framework:

Playbook - Network Reconnaissance (Nmap)

IR Playbook: Reconnaissance Detection

- **Trigger:** Splunk Alert "High Volume Port Scan Detected".
- **Identification:** * Source: Kali Linux (192.168.250.128).
 - Method: TCP SYN Scan (-sS).
- **Response Actions:**
 - Verify if the scan hit internal "Critical Assets" (Team4-DC).
 - Temporary Block: Blacklist the Kali IP on pfSense for 60 minutes.
 - Triage: Determine if this was a precursor to a larger attack.
- **Visual:** A flowchart showing "Detection -> Triage -> Block -> Notify".

Incident Response (IR) Phase

Incident Response Framework:

Playbook - Web Exploitation (SQLi)

IR Playbook: SQL Injection Attempt

- **Trigger:** WAF Alert (ModSecurity) index="waf" showing "UNION SELECT".
- **Identification:** * Target: WEBSRV (192.168.120.100).
 - Impact: Attempted Database credential theft.
- **Response Actions:**
 - **Immediate:** Block Source IP at the WAF level.
 - **Analysis:** Check access.log to see if the HTTP 200 response returned data or just an error.
 - **Remediation:** Set DVWA security level to "Medium/High" or patch the PHP code.
- **Visual:** A screenshot of the ModSecurity "Audit Log" showing the blocked SQL payload.

Incident Response (IR) Phase

Incident Response Framework:

IR Metrics & Mitigation Results

Containment & Recovery Results

- **Mean Time to Detect (MTTD):** < 45 Seconds (from Kali scan start to Splunk alert).
- **Mean Time to Respond (MTTR):** < 3 Minutes (manual IP block on pfSense).
- **Containment Success:** 100% of simulated attacks were isolated to the DMZ; no lateral movement to the Corporate Network was successful.
- **Evidence Preservation:** All logs (Suricata, Sysmon, Apache) successfully indexed in Splunk for forensic review.
- **Visual:** A "Success" checkmark icon next to each of your 5 Use Cases.

SOC Team4



Thank You!

Questions & Live Demo

We can demonstrate live:

- Nmap scan detection in Splunk (pfSense logs)
- SQL Injection payload capture (ModSecurity + Apache)
- Brute force alert trigger (EventCode 4625)

SOC Team 4 — DEPI 2026 | Release 4 | May 2026